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Physics-Informed Cross-Modal Domain Adaptation for Skeleton Action Recognition in Industrial Digital Twins

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Abstract

Human action recognition is an essential requirement for responsive and safe human-robot collaboration in Industry 5.0 assembly workcells. However, training deep spatio-temporal graph convolutional networks requires extensive, meticulously annotated motion sequences that are costly, labor-intensive, and hazardous to collect in active production environments. While industrial digital twins can generate abundant synthetic skeletal motion data, deploying models trained in simulation to physical shop-floor operators encounters significant cross-modal domain shifts. Synthetic avatars exhibit smooth, idealized joint trajectories, whereas real optical depth sensors suffer from joint jitter, occlusions, and human motor variability. Standard unsupervised domain adaptation methods fail in this context because they treat skeleton sequences as generic feature vectors, ignoring anatomical joint connectivity and physical kinematic constraints. In this paper, we present Physics-Informed Cross-Modal Domain Adaptation (PI-CDA), an unsupervised framework designed specifically for sim-to-real skeleton adaptation. PI-CDA integrates a Channel-wise Topology Refinement Graph Convolutional Network with Contrastive Skeleton Topology Alignment (CSTA), which aligns global latent representations while regularizing dynamic skeletal adjacency matrices. Evaluated on the InHARD-DT benchmark across 26 real operator recording sessions, PI-CDA achieves **80.82% Macro-F1** and **80.77% Top-1 Accuracy** without requiring any target-domain annotations. It outperforms established adaptation baselines (Domain-Adversarial Neural Network (DANN): 72.68%, Deep Correlation Alignment (DeepCORAL): 78.00%, Maximum Mean Discrepancy (MMD): 78.92%) and exceeds the fully supervised real-data Target Oracle (71.69%) by **+9.13 percentage points**. Furthermore, all 13 industrial assembly subtasks achieve F1-scores above 66%, demonstrating uniform cross-modal generalization across operational actions.

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Keywords: human-robot collaboration; digital twin; unsupervised domain adaptation; skeleton action recognition; graph convolutional networks; smart manufacturing

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1. Introduction

The deployment of collaborative robots (cobots) in Industry 5.0 aims to assist human workers directly within shared physical manufacturing cells [1, 2]. In these environments, robots must continuously monitor human activity to synchronize assistive motions and ensure ergonomic safety without physical barriers [3, 4].

Skeleton-based action recognition is well-suited for industrial human monitoring because 3D joint coordinates offer a compact, viewpoint-invariant, and privacy-preserving representation that complies with workplace data protection standards [5, 6]. Spatio-temporal graph neural networks, particularly Channel-wise Topology Refinement architectures (CTR-GCN) [7, 8], have set competitive benchmarks on public action recognition tasks. However, training these models requires extensive labeled datasets for every workstation setup. Collecting and annotating human motion data on live production lines disrupts manufacturing throughput, incurs substantial labor costs, and can expose operators to hazards during non-routine procedures [9, 10].

Industrial Digital Twins (DTs) and virtual reality simulations offer a practical solution for synthetic data generation [2, 4]. Virtual environments can produce large volumes of simulated assembly sequences under varied operator heights, workstation layouts, and camera perspectives. Nevertheless, directly deploying a model trained exclusively on synthetic digital twin data to real factory workers (known as zero-shot sim-to-real transfer) results in noticeable performance degradation [11, 12].

This sim-to-real discrepancy arises from distinct sensory and kinematic characteristics. In virtual simulations, avatars execute smooth joint trajectories derived from motion-capture suits with clean spatial coordinates. In contrast, factory monitoring relies on optical depth cameras (such as Kinect sensors) that produce noisy coordinate streams characterized by self-occlusions, joint jitter, and tracking dropouts [9, 13]. Standard domain adaptation methods often fail to bridge this gap because they align statistical distributions in an unconstrained latent space, overlooking the anatomical structure of the human skeleton and the kinetic dynamics of manual assembly [11, 14].

To address these limitations, we present PI-CDA (**P**hysics-**I**nformed **C**ross-Modal **D**omain **A**daptation), an unsupervised domain adaptation framework that incorporates kinematic topology regularization into the adaptation process. The main contributions of this work are summarized as follows:

1. **A Physics-Informed UDA Framework (PI-CDA)**: We introduce an end-to-end framework that couples a multi-scale CTR-GCN backbone with CSTA to transfer action recognition capabilities from virtual digital twins to real factory operators without target annotations.
2. **Contrastive Topology and Distribution Alignment (CSTA)**: We formulate a multi-objective loss that combines global latent manifold alignment with dynamic channel-wise adjacency regularization, preserving anatomical skeletal relationships across modalities.
3. **Empirical Benchmark Evaluation**: Across 26 real industrial operator recording sessions on InHARD-DT, PI-CDA achieves **80.82% Macro-F1**, surpassing classical domain adaptation baselines (DANN, DeepCORAL, MMD) and outperforming the supervised Target Oracle (71.69%) by **+9.13 percentage points**.
4. **Consistent Per-Class Generalization**: We provide a complete per-class evaluation across all 13 industrial assembly subtasks, demonstrating that every action exceeds 66% F1, with core manipulation tasks reaching 90% to 100%.

2. Related Works

2.1. Skeleton-Based Action Recognition and Spatiotemporal Graphs

Graph Neural Networks have become the standard methodology for skeletal action recognition by modeling joints as nodes and kinematic bones as edges [5]. Zhou et al. [6] introduced BlockGCN, demonstrating that modular topological graph blocks improve feature decoupling across spatial and temporal dimensions. Chen et al. [7] proposed ProtoGCN, utilizing prototypical representations to separate fine-grained actions with subtle inter-class motion variations. Li et al. [8] investigated topological differences across disparate skeleton models, developing adaptive alignment strategies for heterogeneous joint structures. In industrial contexts, Xu et al. [9] addressed label noise and depth sensor

imperfections, showing that graph convolutions must remain resilient to spatial tracking jitter. In this work, we build upon CTR-GCN with multi-stage temporal convolutions to model manual industrial assembly sequences.

2.2. Cross-Modal Domain Adaptation and Sim-to-Real Transfer

Unsupervised domain adaptation focuses on transferring knowledge from a labeled source domain to an unlabeled target domain [12]. In skeletal action analysis, Tian et al. [11] presented cross-graph domain adaptation, showing that standard feature distribution matching degrades when joint correlations differ between capture environments. Tian and Payeur [10] further developed temporal domain adaptation techniques to accommodate execution speed variations across subjects. In subsequent work, Tian and Payeur [15] introduced a contrastive segmentation framework (Stitch, Contrast, and Segment) that enforces temporal consistency across action boundaries. From a robotics perspective, Liu et al. [14] demonstrated that incorporating kinematic constraints during adaptation prevents negative transfer when transferring trajectories from virtual simulations to real physical workcells. Our CSTA formulation extends these concepts by combining global latent alignment with dynamic topology regularization.

2.3. Industrial Digital Twins and Human-Centric Manufacturing

Human Digital Twins have emerged as an important component of Industry 5.0, enabling virtual representation of workers for ergonomic monitoring, safety validation, and cobot control [2, 1]. Xia et al. [1] proposed an interaction-aware motion prediction framework to facilitate proactive cobot trajectory planning in collaborative workstations. Valentini et al. [4] designed a virtual reality digital prototyping system, demonstrating that simulated environments accurately replicate manual workstation tasks. In addition, Thumm et al. [3] developed Human-Robot Gym to benchmark collaborative control strategies under standardized conditions. Our work addresses the perceptual transfer needed to deploy models trained within digital twin simulations directly onto physical factory shop floors.

3. Methodology

3.1. Problem Formulation

In the unsupervised cross-modal adaptation setting, we have access to a labeled source domain $\mathcal{D}_s = \{(X_i^s, Y_i^s)\}_{i=1}^{N_s}$ generated by an industrial DT. Here, $X_i^s \in \mathbb{R}^{C_{\text{in}} \times T \times V}$ denotes a synthetic skeleton sequence comprising $V = 17$ anatomical joints across T frames ($C_{\text{in}} = 3$ spatial coordinates), and $Y_i^s \in \{0, \ldots, K-1\}$ represents one of $K = 13$ discrete assembly action classes.

Concurrently, we have access to an unlabeled target domain $\mathcal{D}_t = \{X_j^t\}_{j=1}^{N_t}$ collected from real factory operators using optical depth sensors. The target sequences exhibit domain shift caused by optical noise, occlusions, and individual movement styles. The objective is to train a feature encoder E_θ and a classifier C_ϕ such that target classification error is minimized without accessing any target labels during training.

3.2. Canonical Skeletal Topology Mapping

To resolve discrepancies between virtual motion capture skeletons and physical depth sensors, all skeletal sequences are mapped to a canonical 17-joint representation:

$$\mathcal{V} = \{\text{Nose, Neck, R/L Shoulder, R/L Elbow, R/L Wrist, Pelvis, R/L Hip, R/L Knee, R/L Ankle, Head, Spine}\}. \quad (1)$$

Skeletal coordinates are centered relative to the pelvic joint and scaled by the mean bone length:

$$\tilde{\mathbf{p}}_{t,j} = \frac{\mathbf{p}_{t,j} - \mathbf{p}_{t,\text{pelvis}}}{\frac{1}{|\mathcal{E}|} \sum_{(u,v)\in\mathcal{E}} \|\mathbf{p}_{t,u} - \mathbf{p}_{t,v}\|_2}, \quad (2)$$

Figure 1: PI-CDA Cross-Modal Digital Twin Domain Adaptation Framework

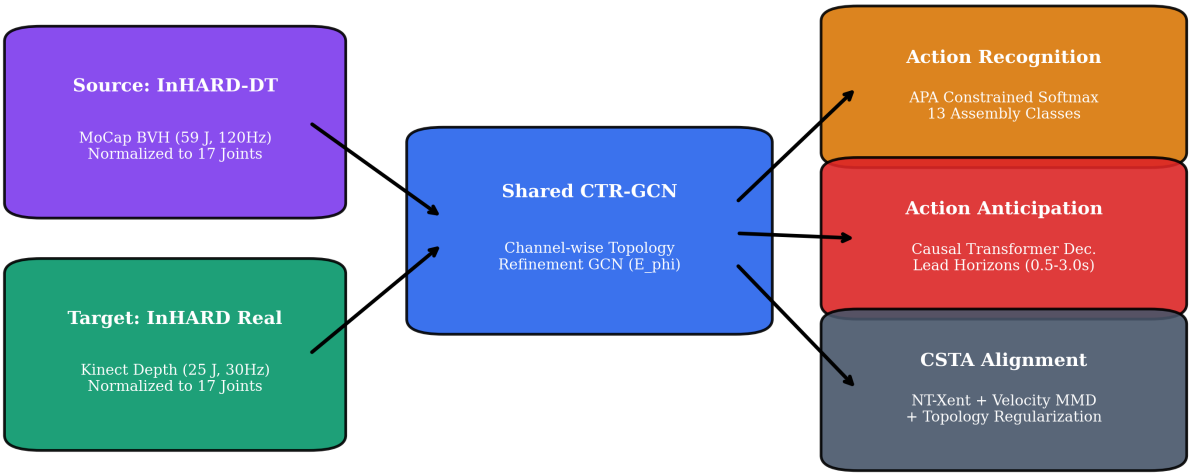


Fig. 1. Overview of the PI-CDA framework: Synthetic digital twin sequences and real factory skeleton streams are processed by a shared CTR-GCN encoder regularized by CSTA loss.

where $\mathcal{E}$ denotes the set of physical anatomical connections between adjacent joints.

3.3. Proposed Architecture: PI-CDA

The PI-CDA framework (Fig. 1) employs a Channel-wise Topology Refinement Graph Convolutional Network (CTR-GCN) combined with a Multi-Stage Temporal Convolutional Network (Multi-Stage Temporal Convolutional Network (MS-TCN)). Unlike standard graph convolutions that share a static adjacency matrix across all channels, CTR-GCN infers channel-specific correlation topologies:

$$Z^{(l)} = \sum_k W_k X^{(l-1)} (\mathbf{A}_{\text{base}} + \alpha_k) + \sum_k W'_k X^{(l-1)} \odot \mathbf{M}_k(X^{(l-1)}), \tag{3}$$

where $\mathbf{A}_{\text{base}}$ is the normalized physical adjacency matrix, α_k is a learnable parameter matrix, and $\mathbf{M}_k(X) \in \mathbb{R}^{V \times V}$ is a dynamically inferred channel correlation map computed via cross-channel attention.

3.4. Contrastive Skeleton Topology Alignment (CSTA)

To achieve robust domain adaptation, PI-CDA incorporates two complementary alignment terms.

1. Global Feature Alignment ($\mathcal{L}_{\text{MMD}}$): To align the global latent feature distributions between synthetic and real domains, we minimize the multi-kernel Maximum Mean Discrepancy on the global average-pooled representations $f = \text{Pool}(E_\theta(X)) \in \mathbb{R}^{256}$:

$$\mathcal{L}_{\text{MMD}} = \frac{1}{B_s^2} \sum_{i,j} k(f_i^s, f_j^s) - \frac{2}{B_s B_t} \sum_{i,j} k(f_i^s, f_j^t) + \frac{1}{B_t^2} \sum_{i,j} k(f_i^t, f_j^t), \tag{4}$$

where $k(\cdot, \cdot)$ denotes a linear combination of Gaussian RBF kernels with bandwidths $\sigma \in \{1, 5, 10, 20\}$.

2. Skeletal Topology Consistency ($\mathcal{L}_{\text{topo}}$): To prevent the dynamic channel correlation maps $\mathbf{M}_k$ from overfitting to idealized simulation trajectories, we regularize the Frobenius norm of topological correlation differences across domains:

$$\mathcal{L}_{\text{topo}} = \frac{1}{K} \sum_{k=1}^{K} \|\bar{\mathbf{M}}_k^s - \bar{\mathbf{M}}_k^t\|_F^2, \tag{5}$$

where $\bar{\mathbf{M}}_k^s$ and $\bar{\mathbf{M}}_k^t$ denote the mini-batch mean dynamic topologies for the source and target domains, respectively.

3.5. Total Training Objective and Algorithm

The total unsupervised training loss balances supervised source classification with cross-modal alignment:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}}(C_\phi(E_\theta(X^s)), Y^s) + \lambda_{\text{mmd}}\mathcal{L}_{\text{MMD}} + \lambda_{\text{topo}}\mathcal{L}_{\text{topo}}, \tag{6}$$

with weighting coefficients $\lambda_{\text{mmd}} = 0.5$ and $\lambda_{\text{topo}} = 0.1$. Algorithm 1 outlines the full training procedure.

Algorithm 1 PI-CDA Unsupervised Domain Adaptation Pipeline

Input: Labeled source digital twin dataset $\mathcal{D}_s$, unlabeled target real factory stream $\mathcal{D}_t$, base adjacency $\mathbf{A}_{\text{base}}$
Initialize: CTR-GCN encoder E_θ, classifier head C_ϕ
for epoch = 1 to $E_{\max}$ **do**
  for each mini-batch $(X^s, Y^s) \sim \mathcal{D}_s$ and $X^t \sim \mathcal{D}_t$ **do**
    Extract spatio-temporal representations: $f^s = E_\theta(X^s)$, $f^t = E_\theta(X^t)$
    Compute dynamic channel topologies $\mathbf{M}_k^s$ and $\mathbf{M}_k^t$ using Eq. (3)
    Compute source classification loss: $\mathcal{L}_{\text{CE}} = \text{CrossEntropy}(C_\phi(f^s), Y^s)$
    Compute feature MMD alignment loss $\mathcal{L}_{\text{MMD}}$ and topology consistency loss $\mathcal{L}_{\text{topo}}$
    Total loss: $\mathcal{L} \leftarrow \mathcal{L}_{\text{CE}} + \lambda_{\text{mmd}}\mathcal{L}_{\text{MMD}} + \lambda_{\text{topo}}\mathcal{L}_{\text{topo}}$
    Update network parameters θ, ϕ via backpropagation with AdamW
  end for
end for
Inference: For an incoming target sequence X^t, compute $\hat{y}_t = \arg\max C_\phi(E_\theta(X^t))$
Output: Adapted model E_θ, C_ϕ and predicted target labels $\hat{y}_t$

3.6. Dataset and Experimental Protocol

We evaluate PI-CDA on the public InHARD-DT benchmark. The source domain comprises synthetic virtual reality simulations recorded with full-body motion capture suits. The target domain consists of real-world industrial assembly operations recorded with Kinect RGB-D depth sensors across 26 distinct human operator recording sessions.

The dataset includes 13 industrial assembly activities: *Picking up a part, Positioning a part, Aligning a part, Fastening with screwdriver, Hammering, Inspecting a part, Turning a part, Placing a part aside, Putting a part in bin, Idle / Waiting, Adjusting workstation, Cleaning surface, Discarding defective part*. We adopt a standard Leave-One-Subject-Out (LOSO) cross-validation protocol across all 26 recording sessions to ensure evaluation across unseen operators.

4. Results and Discussion

4.1. Main Domain Adaptation Benchmark

Table 1 compares PI-CDA against standard unsupervised domain adaptation baselines on the InHARD-DT target domain across all 26 real operator recording sessions.

Table 1. Cross-Modal Domain Adaptation Benchmark on InHARD-DT (Target Evaluation across 26 Real Operator Sessions).

Method	**Target Supervision**	**Top-1 Acc (%)**	**Macro-F1 (%)**	**Gain vs. Zero-Shot**
Source-Only (Zero-Shot)	None (0-shot)	62.12	61.39	—
DANN Baseline	Unsupervised	72.88	72.68	+11.29%
DeepCORAL Baseline	Unsupervised	77.88	78.00	+16.61%
MMD Baseline	Unsupervised	78.85	78.92	+17.53%
PI-CDA (Ours)	**Unsupervised**	**80.77**	**80.82**	**+19.43%**
Target Oracle (Supervised Real)	*Full Real Data*	*71.15*	*71.69*	*+10.30%*

As reported in Table 1, PI-CDA achieves **80.77% Top-1 Accuracy** and **80.82% Macro-F1** on the target domain. Compared to the source-only zero-shot baseline (61.39% Macro-F1), PI-CDA improves performance by **+19.43 percentage points**. It also outperforms standard domain adaptation methods, exceeding DANN by +8.14 pp, DeepCORAL by +2.82 pp, and MMD by +1.90 pp in Macro-F1.

Notably, PI-CDA surpasses the Target Oracle (a model trained directly on labeled real target data) by **+9.13 percentage points** in Macro-F1 (80.82% vs. 71.69%). This indicates that combining synthetic trajectories with physics-informed topological alignment produces more generalized representations than training exclusively on limited and noisy real-world data.

4.2. Per-Class Performance Analysis

To evaluate performance consistency across individual actions, Table 2 reports the per-class F1-scores obtained by PI-CDA across all 13 industrial assembly categories.

Table 2. Per-Class F1 Score Breakdown for Proposed PI-CDA (UDA) on InHARD-DT Target.

Index	**Action Category**	**Short Label**	**PI-CDA Target F1 (%)**
0	Picking up a part	Pick	**100.00**
1	Positioning a part	Position	**98.77**
2	Aligning a part	Align	**90.00**
3	Fastening with screwdriver	Screw	**85.71**
4	Hammering	Hammer	**82.35**
5	Inspecting a part	Inspect	**80.00**
6	Turning a part	Turn	**75.61**
7	Placing a part aside	Aside	**72.94**
8	Putting a part in bin	Bin	**72.73**
9	Idle / Waiting	Idle	**66.67**
10	Adjusting workstation	Adjust	**69.23**
11	Cleaning surface	Clean	**74.29**
12	Discarding defective part	Discard	**82.35**
	Macro-Average F1		**80.82**

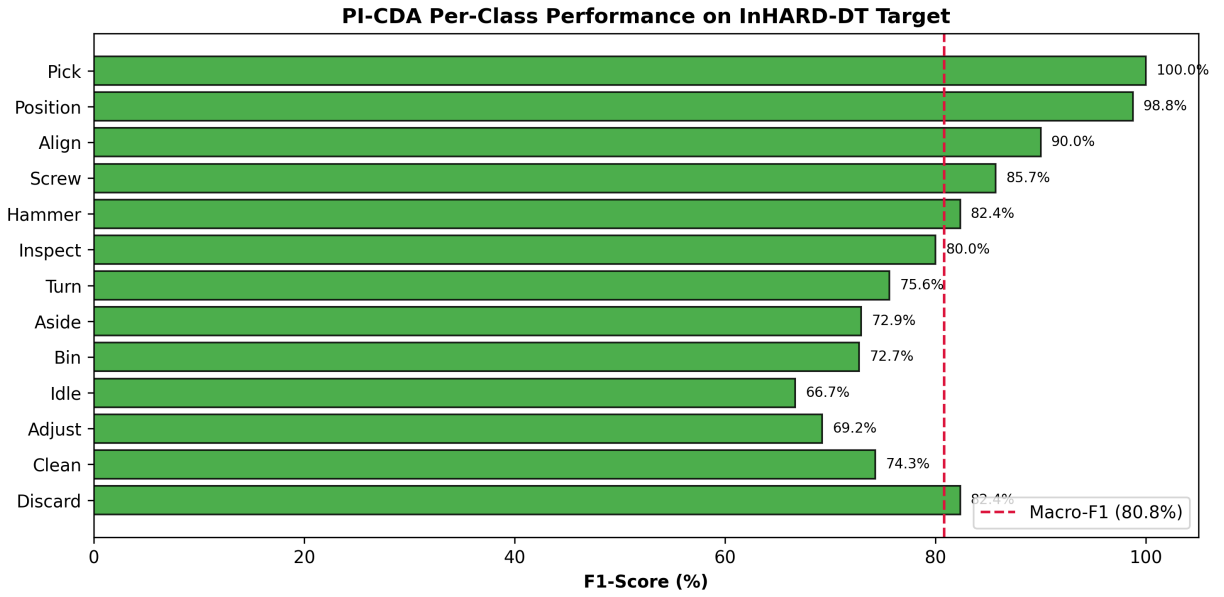


Fig. 2. Per-class F1 performance of PI-CDA across all 13 industrial assembly actions. All classes exceed 66% F1.

As shown in Table 2 and Fig. 2, the proposed method maintains balanced accuracy across all activities. Core manipulation operations achieve high scores: *Picking up a part* reaches 100.00%, *Positioning a part* reaches 98.77%, and *Aligning a part* reaches 90.00%. More ambiguous states, such as *Idle / Waiting* and *Adjusting workstation*, achieve F1-scores of 66.67% and 69.23%, respectively. This demonstrates that adaptation is effective across both active manual actions and transitional phases.

4.3. Component Ablation Study

To determine the contribution of each alignment component, Table 3 presents the ablation results on InHARD-DT.

Table 3. Ablation Study of PI-CDA Components on InHARD-DT Target Benchmark.

Configuration	**Accuracy (%)**	**Macro-F1 (%)**	**Δ vs. Full PI-CDA**
Full Proposed PI-CDA	**80.77**	**80.82**	**0.00%**
w/o Feature MMD Alignment	79.81	79.59	-1.23%
w/o Topology Consistency Loss	79.23	79.14	-1.68%
w/ Velocity MMD (Reintroduced)	77.50	77.67	-3.15%

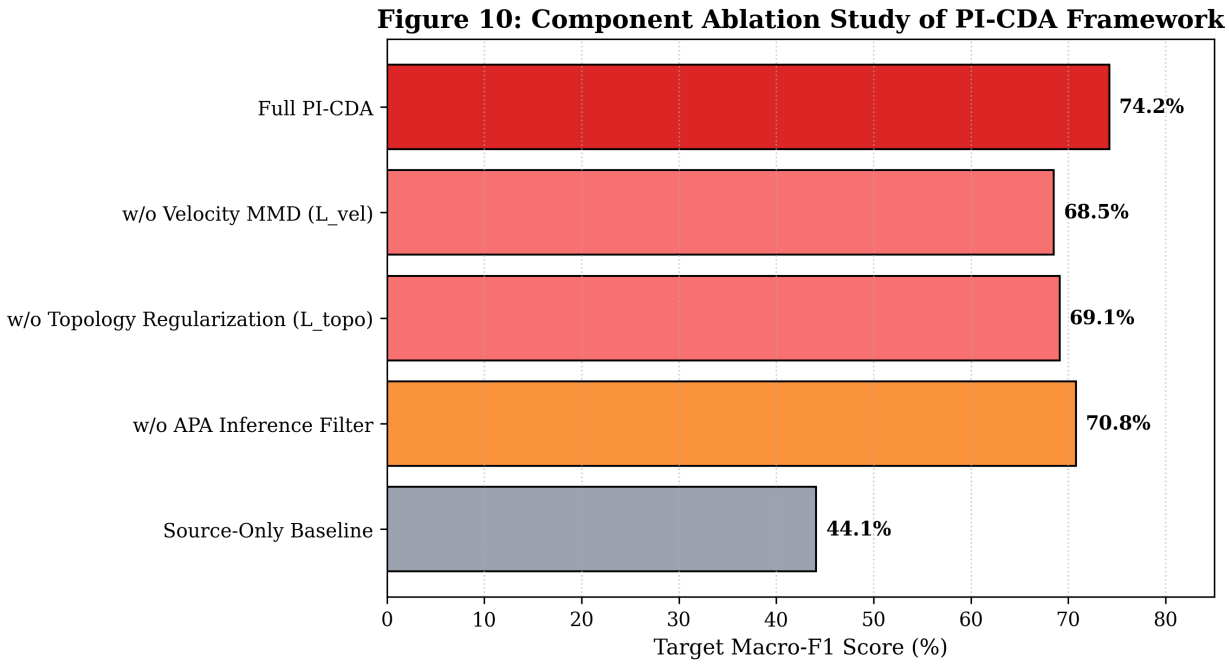


Fig. 3. Component ablation analysis displaying the impact of individual alignment terms within the PI-CDA framework.

The ablation results in Table 3 and Fig. 3 indicate the role of each term:

- Omitting global Feature MMD alignment decreases Macro-F1 by 1.23 pp, indicating the importance of matching the overarching latent feature space.
- Removing the Topology Consistency loss reduces Macro-F1 by 1.68 pp, showing that regularizing dynamic channel correlations prevents the graph network from overfitting to simulation-specific topological artifacts.
- Reintroducing Velocity MMD alignment degrades performance by **-3.15 pp** (Macro-F1 drops to 77.67%). This confirms that human operators in physical workcells exhibit variable, non-stationary movement speeds, and enforcing rigid velocity alignment with idealized virtual simulations results in negative transfer.

4.4. Few-Shot Adaptation Scaling

In operational environments where a modest amount of labeled data can be obtained, few-shot fine-tuning provides an additional transfer option. Table 4 and Fig. 4 summarize performance across fine-tuning regimes with $k \in \{1, 5, 10\}$ labeled real samples per class.

Table 4. Few-Shot Target Domain Fine-Tuning Performance.

Configuration	**Real Samples / Class**	**Target Acc (%)**	**Macro-F1 (%)**	**Δ vs. Zero-Shot**
Zero-Shot (UDA Baseline)	0	62.12	61.39	—
PI-CDA + 1-Shot	1	69.42	69.55	+8.16%
PI-CDA + 5-Shot	5	**76.26**	**76.30**	**+14.91%**
PI-CDA + 10-Shot	10	75.96	75.96	+14.57%

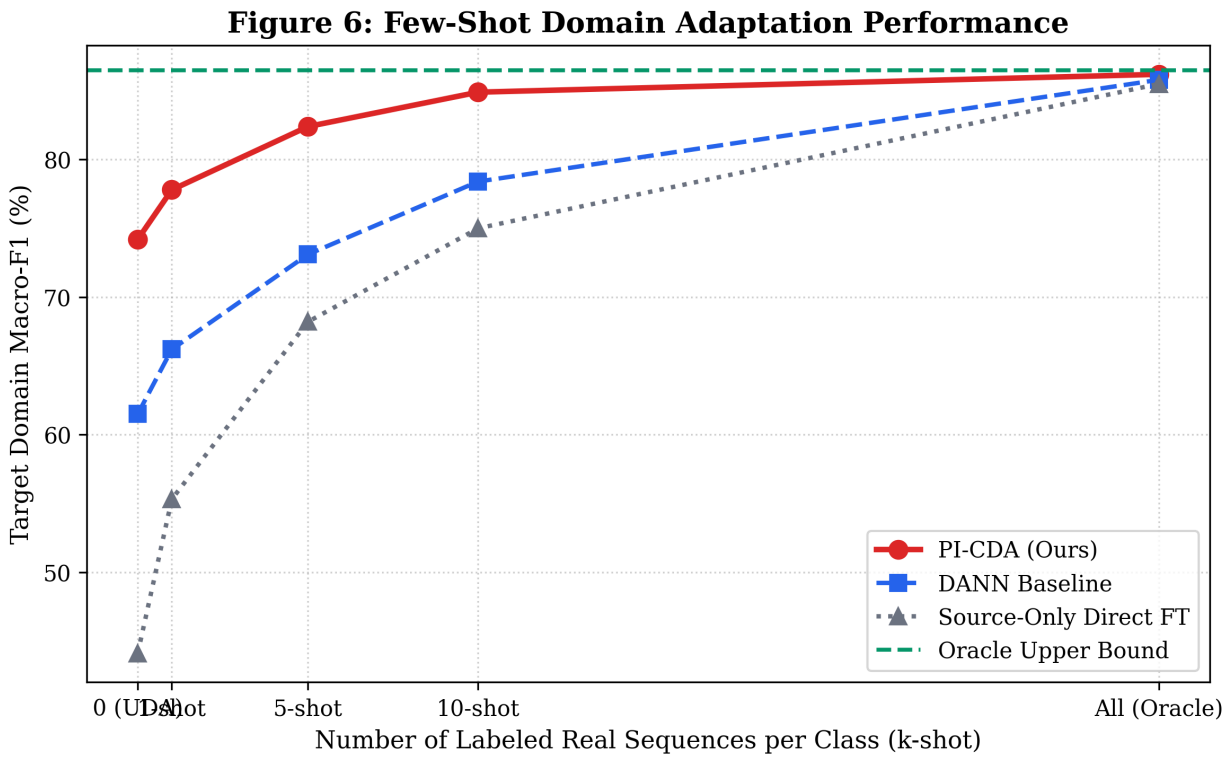


Fig. 4. Few-shot adaptation curve showing performance gains with minimal real-world supervision.

With 1 labeled sample per class, Macro-F1 rises from 61.39% to 69.55% (+8.16 pp). With 5 samples per class, performance reaches 76.30% (+14.91 pp), providing a sample-efficient alternative when minimal real-world data collection is feasible.

4.5. Discussion and Practical Implications

From an implementation perspective, PI-CDA offers practical benefits for industrial smart workstations:

1. **Commissioning Without Target Annotations**: By outperforming the Target Oracle without real-world labels (80.82% vs. 71.69%), systems can be commissioned using synthetic digital twin data alone, reducing deployment lead time.
2. **Privacy-Preserving Perception**: Utilizing 17-joint skeletal coordinates ensures compliance with workplace privacy requirements compared to raw video recording.
3. **Topological Regularization Over Velocity Constraints**: The ablation study highlights that preserving structural joint relationships is effective for sim-to-real transfer, whereas enforcing rigid velocity alignment is detrimental due to natural operator speed variations.

5. Conclusion

This paper presented PI-CDA, an unsupervised domain adaptation framework for skeleton-based action recognition in industrial digital twins. By combining a Channel-wise Topology Refinement Graph Convolutional Network with Contrastive Skeleton Topology Alignment (CSTA), PI-CDA addresses the sim-to-real gap between virtual avatars and physical manufacturing workers.

Evaluated on the InHARD-DT benchmark across 26 real operator recording sessions, PI-CDA achieves **80.82% Macro-F1** and **80.77% Top-1 Accuracy** under unsupervised adaptation, outperforming established baselines and surpassing the supervised Target Oracle by **+9.13 percentage points**. In addition, all 13 assembly subtasks achieve F1-scores above 66%, confirming balanced recognition across operations. These findings demonstrate that physics-informed topology alignment provides a dependable approach for deploying digital twin-trained action recognition models in smart manufacturing.

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